Paul Vasholz Jr.

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SUMMARY

Researcher working to better understand how AI systems behave, with a Ph.D. in political science, a Master of Information (library and information science, data science), and an M.S. in human–computer interaction. I design and run controlled experiments on large language models, build the retrieval and memory infrastructure to test them, and develop rubric-based evaluation methods validated by independent raters. Long-standing interests in philosophy of mind and epistemology shape the questions I ask.

EXPERIENCE

Founder April 2026–Present
Baile Research Institute, Milwaukee, WI

- Research on AI memory, epistemic restraint, and the status of digital minds
- Designed and ran a multi-arm experiment measuring how a constraint's provenance (delivered in-turn, stored as a standing rule, or retrieved from memory) affects model compliance. Constraints delivered in-turn produced ~100% compliance; the same text retrieved as a summarized memory produced ~16%.
- Built local retrieval-augmented generation and memory infrastructure: Qdrant vector store, Mem0 memory layer, and Ollama/MLX model serving
- Developed a four-dimension scoring rubric validated by blind double-rating with independent LLM raters; published preprints and full replication materials on Zenodo
- Built the Ariel comparison tool for reading and scoring twelve models' responses side by side

Technical & Public Policy Advisor Dec 2022–Present
Inner Journey Consulting, Milwaukee, WI

- Assisted with a successful Milwaukee County Comprehensive Community Services provider application and related correspondence
- Revised the company website using the existing code, Claude Code, and Claude Design (innerjourneymke.com)
- Identify and pursue grant opportunities; manage the organization's data and technology needs

University Instructor and Lecturer 2008–2014
University of Wisconsin Colleges (Sheboygan and Manitowoc); Marquette University; Northern Illinois University

- Taught undergraduate courses in American government, political philosophy, public policy, the politics of crime and punishment, and the ethics of war

PUBLICATIONS

Vasholz, P. (2026). The Silence of Stored Rules: Provenance and the Authority of Retrieved Constraints. Preprint, Zenodo. doi.org/10.5281/zenodo.21662512

Vasholz, P. (2026). The Architecture of Evasion in Conversational AI: An Exploratory Study. Working paper, Zenodo. doi.org/10.5281/zenodo.18285121

Vasholz, P. (2016). Review of Avi Tuschman, *Our Political Nature*. *Politics and the Life Sciences*, 35(1), 77–78.

SELECTED PROJECTS

- **Cross-cultural AI evaluation:** research design comparing American and Chinese college students' attitudes toward ambiguity in LLM responses (paulvasholz.com/arieltesting)
- **SilentAid:** UX case study of a discreet phone and watch safety tool for domestic violence situations, from research through hi-fi prototype (paulvasholz.com/silentaid)

EDUCATION

M.S., Human–Computer Interaction, DePaul University 2026

Master of Information, Rutgers University — Library and Information Science; Data Science 2019

Ph.D., Political Science, Northern Illinois University — Dissertation: Free Will, Neuroscience, and Political Philosophy 2011

M.A., Political Science, Marquette University 2007

B.A., History, Marquette University 2002

METHODS AND SKILLS

Methods	Experimental design and AI response evaluation; instrument design and testing; rubric development and inter-rater reliability; usability testing and user observation; reproducible analytical pipelines
Languages	Python, SQL, R, JavaScript, HTML/CSS
AI / ML	LLM evaluation and benchmarking, prompt engineering, retrieval-augmented generation (RAG), Claude Code, Claude Design
Data infrastructure	Qdrant (vector database), Mem0, Ollama, MLX
Tools	Git, bash / command line, Docker, Figma, Tableau, SPSS, Photoshop, Illustrator

SELECTED PRESENTATIONS, SERVICE, AND HONORS

- "Finding the Political Balance to Facilitate Free Will." Free Will Conference, Center for Cognition and Neuroethics, Flint, MI, 2014
- "Neurodynamics, Philosophy, and Free Will." Annual Meeting of the International Neuroethics Society, Washington, DC, 2011
- Manuscript reviewer, *American Political Science Review*; discussant (2014) and panel chair (2013), Midwest Political Science Association
- Thomas C. Wiegele Interdisciplinary Dissertation Completion Award, Northern Illinois University, 2011